

```

Clear["Global`*"];
h1 = 100;
h2 = 200;
n1 = 1.5;
n2 = 2.5;
hd = 300;
Nbef = 7;
Naft = 7;
n0 = 1;
n01 = 1;
ε = 1;
μ = 1;
θ = 0;
p1 = n1 * Cos[θ];
k1 = (2 * Pi / lambda) * n1 * Cos[θ];
m1 =
  {{Cos[k1 * h1], (-I / p1) * Sin[k1 * h1]}, {-I * p1 * Sin[k1 * h1], Cos[k1 * h1]}};
p2 = n2 * Cos[θ];
k2 = (2 * Pi / lambda) * n2 * Cos[θ];
m2 =
  {{Cos[k2 * h2], (-I / p2) * Sin[k2 * h2]}, {-I * p2 * Sin[k2 * h2], Cos[k2 * h2]}};
pd = nd * Cos[θ];
kd = (2 * Pi / lambda) * nd * Cos[θ];
md =
  {{Cos[kd * hd], (-I / pd) * Sin[kd * hd]}, {-I * pd * Sin[kd * hd], Cos[kd * hd]}};
p0 = 1; pf = 1;
m3 = MatrixPower[m1.m2, Nbef].md.MatrixPower[m1.m2, Naft];
t = (2 * p0) / ((m3[[1, 1]] + pf * m3[[1, 2]]) * p0 + (m3[[2, 1]] + pf * m3[[2, 2]]));
r = ((m3[[1, 1]] + pf * m3[[1, 2]]) * p0 - (m3[[2, 1]] + pf * m3[[2, 2]])) /
  ((m3[[1, 1]] + pf * m3[[1, 2]]) * p0 + (m3[[2, 1]] + pf * m3[[2, 2]]));
T = (pf / p0) * Abs[t^2];
R = Abs[r^2];
center1 = 2 * (n1 * h1 + n2 * h2)
center2 = (n1 * h1 + n2 * h2)
Plot[T /. nd → 1.33, {lambda, 0, 1800}, PlotRange → All]
Plot[R /. nd → 1.33, {lambda, 0, 1800}, PlotRange → All]

(*3rd zone*)
Max1 = FindMaximum[{T /. nd → 1.33, 447 ≤ lambda ≤ 451}, {lambda, 448}]
Max2 = FindMaximum[{T /. nd → 1.331, 447 ≤ lambda ≤ 451}, {lambda, 448}]
(*sensitivity*)
(Abs[Max1[[2, 1, 2]] - Max2[[2, 1, 2]]]) / (0.001)
(*normalized sensitivity*)
(Abs[Max1[[2, 1, 2]] - Max2[[2, 1, 2]]]) / (0.001 * 449.2709401287054)

(*2nd zone*)

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Max3 = FindMaximum[{T /. nd → 1.33, 610 ≤ lambda ≤ 620}, {lambda, 618}]
Max4 = FindMaximum[{T /. nd → 1.331, 610 ≤ lambda ≤ 620}, {lambda, 618}]
(*sensitivity*)
(Abs[Max3[[2, 1, 2]] - Max4[[2, 1, 2]]]) / (0.001)
(*normalized sensitivity*)
(Abs[Max3[[2, 1, 2]] - Max4[[2, 1, 2]]]) / (0.001 * 618.6134003150806)

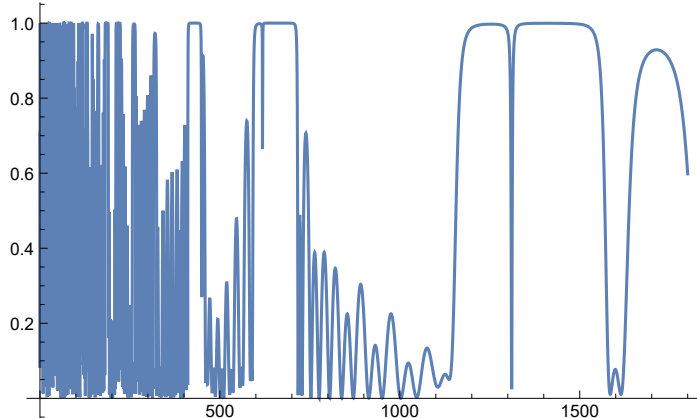
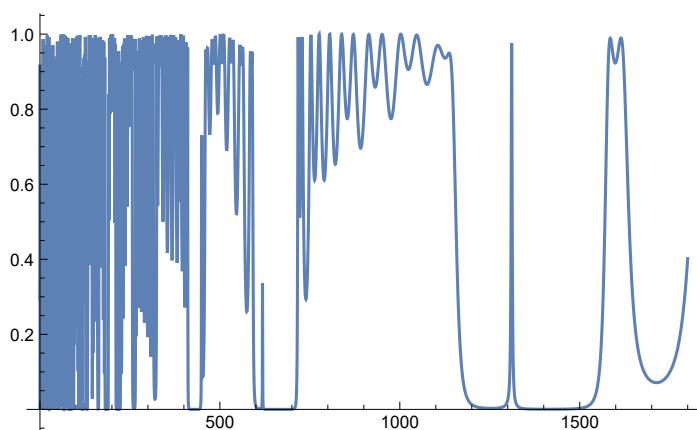
(*1st zone*)
Max5 = FindMaximum[{T /. nd → 1.33, 1300 ≤ lambda ≤ 1320}, {lambda, 1310}]
Max6 = FindMaximum[{T /. nd → 1.331, 1300 ≤ lambda ≤ 1320}, {lambda, 1310}]
(*sensitivity*)
(Abs[Max5[[2, 1, 2]] - Max6[[2, 1, 2]]]) / (0.001)
(*normalized sensitivity*)
(Abs[Max5[[2, 1, 2]] - Max6[[2, 1, 2]]]) / (0.001 * 1310.8236599221318)

(*3rd zone*)
Min1 = FindMinimum[{R /. nd → 1.33, 447 ≤ lambda ≤ 451}, {lambda, 448}]
Min2 = FindMinimum[{R /. nd → 1.331, 447 ≤ lambda ≤ 451}, {lambda, 448}]
(*sensitivity*)
(Abs[Min1[[2, 1, 2]] - Min2[[2, 1, 2]]]) / (0.001)
(*normalized sensitivity*)
(Abs[Min1[[2, 1, 2]] - Min2[[2, 1, 2]]]) / (0.001 * 449.2709401287371)

(*2nd zone*)
Min3 = FindMinimum[{R /. nd → 1.33, 610 ≤ lambda ≤ 620}, {lambda, 618}]
Min4 = FindMinimum[{R /. nd → 1.331, 610 ≤ lambda ≤ 620}, {lambda, 618}]
(*sensitivity*)
(Abs[Min3[[2, 1, 2]] - Min4[[2, 1, 2]]]) / (0.001)
(*normalized sensitivity*)
(Abs[Min3[[2, 1, 2]] - Min4[[2, 1, 2]]]) / (0.001 * 618.6134003150693)

(*1st zone*)
Min5 = FindMinimum[{R /. nd → 1.33, 1300 ≤ lambda ≤ 1320}, {lambda, 1310}]
Min6 = FindMinimum[{R /. nd → 1.331, 1300 ≤ lambda ≤ 1320}, {lambda, 1310}]
(*sensitivity*)
(Abs[Min5[[2, 1, 2]] - Min6[[2, 1, 2]]]) / (0.001)
(*normalized sensitivity*)
(Abs[Min5[[2, 1, 2]] - Min6[[2, 1, 2]]]) / (0.001 * 1310.8236599218005)
1300.
650.

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{0.967969, {lambda → 449.271}}

{0.968059, {lambda → 449.305}}

34.3443

0.0764446

{0.952352, {lambda → 618.613}}

{0.951911, {lambda → 618.699}}

85.6846

0.138511

{0.975601, {lambda → 1310.82}}

{0.975522, {lambda → 1311.07}}

246.963

0.188403

{0.0320308, {lambda → 449.271}}

{0.0319412, {lambda → 449.305}}

34.3443

0.0764446

{0.0476485, {lambda → 618.613}}

$\{0.048089, \{\lambda \rightarrow 618.699\}\}$

85.6846

0.138511

$\{0.0243989, \{\lambda \rightarrow 1310.82\}\}$

$\{0.0244783, \{\lambda \rightarrow 1311.07\}\}$

246.963

0.188403